

GENERAL SPECIFICATION G2.25

Manufacturers' Services

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G2.25 MANUFACTURERS' SERVICES

G2.25.1 Definitions

- (a) Products: Includes material, equipment, systems or sub-systems, cast-in or embedded items or components provided by the Contractor and installed under the Contract or handed over to the Board as spares or accessories.

G2.25.2 Qualification of Manufacturers' Representatives

- (a) The authorised representatives of the manufacturers shall be proficient in English. They shall be factory trained and experienced in the technical applications, installation, commissioning, operation, and maintenance of respective equipment, system, or subsystem, with full authority by the equipment manufacturers to issue the certifications required of the manufacturers. Additional qualifications may be specified elsewhere.
- (b) The Contractor shall note that manufacturers' representatives will be subjected to acceptance by the S.O.. Substitution of representatives shall not be allowed unless prior written approval by S.O. has been given.

G2.25.3 Fulfilment of Specified Minimum Services

- (a) The Contractor shall furnish manufacturers' services when required by an individual Specification Section to meet the requirements of this Section.
- (b) Where additional time is needed in excess of that stated in the Specifications for provision of manufacturers' services, or when a minimum time is not specified, the time and cost required to perform the specified services shall deemed to be included in the Contract Sum.
- (c) The Contractor shall programme manufacturers' services to avoid conflict with other onsite testing or other manufacturers' onsite services.
- (d) The Contractor shall determine, before programming manufacturers' services, that all conditions necessary to allow successful testing have been met.
- (e) The Contractor shall note that only those days of service approved by the S.O. will be credited to fulfil the specified or agreed minimum services.
- (f) When specified in individual Specification sections, manufacturers' services shall include, but shall not be limited to:
 - (i) Assistance during product (system, subsystem, or component) installation to include observation, guidance, instruction to Contractor for the proper erection, installation or application procedures;
 - (ii) Inspection, checking, and adjustment as required for product (system, subsystem, or component) to function as warranted by manufacturers and necessary to furnish Manufacturers' Certificate of Proper Installation;
 - (iii) Providing, on a daily basis, copies of all manufacturers' representatives' field notes and data to S.O.;
 - (iv) Revisiting the Site as required to correct problems until installation and operation are acceptable to S.O.;

- (v) Resolution of assembly or installation problems attributable to, or associated with, respective manufacturers' products and systems; and
- (vi) Additional requirements may be specified elsewhere.

G2.25.4 Manufacturers' Test Reports and Certificates of Compliance

- (a) The Contractor shall submit the factory test reports and Certificates of Compliance for products specified to a recognised standard or code prior to shipment of products or materials. The test reports and Certificates of Compliance shall be furnished with each lot of products or materials delivered to the site and such test reports and Certificates of Compliance shall identify clearly the lot of products or materials so certified.
- (b) The S.O. may permit use of certain materials or assemblies prior to field-testing if accompanied by accepted Certificates of Compliance. The S.O. reserves the right to refuse permission for use of products on the basis of the Certificates of Compliance.
- (c) All manufacturers' test reports and certificates of compliance shall be signed by product manufacturers certifying that the products or materials specified conform to or exceed the specifications. Supporting reference data, affidavits, and certifications as appropriate shall be attached.

G2.25.5 Manufacturers' Designs

- (a) Any designs by manufacturers shall be in accordance with the provisions of the latest edition of the Building Control Act, Singapore Standards, British Standards, American Standards or other recognised Standards, where applicable. These designs shall be performed and endorsed by the Contractor's Professional Engineers. Where required under the Building Control Act, the Contractor shall arrange for the designs to be checked by Accredited Checkers prior to submission to the Building and Construction Authority. All such costs shall be deemed included in the rates.

G2.25.6 Manufacturers' Certificates of Proper Installation

- (a) Where specified, the Contractor shall furnish certification stating the following as a minimum:
 - (i) The product or system has been installed in accordance with the manufacturers' recommendation;
 - (ii) The product or system has been inspected by the manufacturers' authorised representatives;
 - (iii) The product or system has been serviced with proper lubricants;
 - (iv) Applicable safety equipment has been properly installed; and
 - (v) Proper adjustments have been made and the product or system is ready for site testing, performance testing and operation.

- (b) A copy of the Manufacturer's Certificate of Proper Installation form is appended to this Section. This form shall be completed and signed by the system or equipment manufacturers' representatives and the Contractor. Such forms shall certify that the signing party is a duly authorised representative of the manufacturer, is empowered by the manufacturer to inspect, approve, and operate their system or equipment and is authorised to make recommendations required to assure that the material, system or equipment is complete and operational.

G2.25.7 Submittals

- (a) The Contractor shall submit the following under this Section, and as specified in the general specifications:
 - (i) Qualification of Manufacturers' Representatives. The Contractor shall submit resumes at least 30 days prior to commencement of work, where specified for specific equipment or product; and
 - (ii) Manufacturers' Certificates of Proper Installation.

G2.25.8 Appendix

- (a) APPENDIX G2.25A - Manufacturer's Certificate of Proper Installation.

END OF SECTION

APPENDIX G2.25A
MANUFACTURER'S CERTIFICATE OF PROPER INSTALLATION

OWNER: Public Utilities Board, Singapore

CERTIFICATE NO: _____

Project name: Tuas Water Reclamation Plant
Contract CXX

Equipment TAG NO: _____

EQPT/SYSTEM: _____

I hereby certify that the above-referenced equipment/system has been:

(Check Where Applicable)

- ☐ Installed in accordance with Manufacturer's recommendations.
- ☐ Inspected, checked, and adjusted.
- ☐ Serviced with proper initial lubricants.
- ☐ Electrical and mechanical connections meet quality and safety standards.
- ☐ All applicable safety equipment has been properly installed.
- ☐ Proper adjustments have been made and the product or system is ready for functional testing, performance testing and operation.

Comments:

I, the undersigned Manufacturer's Representative, hereby certify that I am (i) a duly authorised representative of the manufacturer, (ii) empowered by the manufacturer to inspect, approve, and operate his system/equipment and (iii) authorised to make recommendations required to assure that the system/equipment furnished by the manufacturer is complete and operational, except as may be otherwise indicated herein. I further certify that all information contained herein is true and accurate.

Date:

Manufacturer:

By Manufacturer's Authorised Representative:

(Name)

(Authorised Signature)